

Prompt Engineering in Fine-Tuning BiomedCLIP to Enhance COVID-19 Chest X-Ray Diagnosis

Aochen He

The School of Computing and Data Science

The University of Hong Kong

Hong Kong, China

hac1224@connect.hku.hk

Abstract—Large-scale pre-trained Vision-Language Models (VLMs), such as BiomedCLIP, have demonstrated promising potential for biological image classification. However, effectively adapting these large pre-trained VLMs to specialized diagnostic tasks remains challenging due to the scarcity of annotated datasets and the prohibitive computational costs associated with full-model fine-tuning. Prompt engineering, which aims to rapidly adapt pre-trained models on small-scale, task-specific datasets, has emerged as a widely adopted alternative. Nevertheless, current prompt engineering methodologies lack standardization. To address this limitation, this work proposes a novel diagnostic axis framework. It standardizes prompts to systematically capture clinically relevant features while accounting for model sensitivity to stylistic prompt variations. This approach enhances both task adaptation and prompt interpretability for downstream diagnostic applications. On the COVID-19 Radiography Database, the model fine-tuned under the proposed framework achieved a 20.33% improvement in Accuracy compared to the baseline model. Experimental results demonstrate that models fine-tuned using diagnostic axes significantly outperformed zero-shot baseline models across most evaluation metrics on a COVID-19 screening task. Despite these promising results, this study has several limitations, such as insufficient exploration of interactions among different prompt styles and limited support for complex reasoning tasks. Future research could further incorporate potential interactions and extend support for multi-step reasoning to improve the generalizability and practical utility of the method.

Keywords—Prompt Engineering, BiomedCLIP, COVID-19

I. INTRODUCTION

The COVID-19 pandemic has resulted in millions of deaths worldwide, and its recurring outbreaks, along with the continuous emergence of new variants, have posed a profound global public health threat. In this context, artificial intelligence-assisted diagnostic technologies, particularly rapid screening tools based on medical imaging, play a crucial role in pandemic control. Currently, Convolutional Neural Networks (CNNs), as a representative architecture in the field of deep learning, have demonstrated remarkable performance in the classification of COVID-19 chest X-ray images, with some studies reporting accuracy rates as high as 99.7% [1]. However, such models still face three fundamental challenges in real-world clinical deployment: first, the scarcity of high-quality medical imaging data leads to insufficient training samples; second, complex and variable data distribution shifts in practical scenarios significantly reduce model robustness; and third, the model's generalization capability for downstream derived tasks remains considerably limited. These bottlenecks

severely hinder the translation of CNN models from laboratory research to bedside application [2].

To address these limitations, Visual Language Models (VLMs) have become a new focus in medical image analysis. The revolutionary value of these models lies in their ability to learn both visual representations and semantic textual information simultaneously. The dual learning capability enables them to perform complex tasks such as image captioning, visual question answering, and multimodal retrieval [3]. It is particularly noteworthy that the success of the Contrastive Language-Image Pre-training (CLIP) model has demonstrated the decisive guiding role of text supervision in visual representation learning. This finding has directly spurred the exploration of VLMs in medical imaging [4]. For instance, some researchers introduce multi-scale contrastive learning methods during pre-training to enhance CLIP's performance in object detection tasks, and other approaches utilize semantic correlation or mining auxiliary information to improve pre-training efficiency and representation power [5-7]. Although VLMs demonstrate excellent cross-domain transferability and general-purpose representation capabilities, effectively applying them to meet the specific needs of medical scenarios remains a challenge. Traditional fine-tuning strategies, such as full parameter updating, face limitations in medical applications. They not only demand large amounts of annotated data, but also often fail to fully activate the inherent medical knowledge within the models.

As these technologies evolve, prompt engineering has become an important way to extract medical usefulness from pre-trained models. It works by feeding the model carefully written prompts that nudge it to express the proper knowledge, allowing it to fit a specialized domain without much retraining. This semantic modulation mechanism provides pre-trained models with fine-grained and low-cost domain adaptation capabilities [2]. For example, incorporating expert domain knowledge into descriptive text prompts enhances classification supervision for medical data [8]. Other approaches employ multiple prompts to characterize medical lesions, enabling zero-shot lesion detection [7]. However, existing prompt techniques still suffer from systematic shortcomings: first, prompt design heavily relies on researchers' experiential intuition and lacks standardized design principles; second, the optimization process lacks quantifiable evaluation metrics; and third, the decision logic of prompt strategies exhibits a black-box nature, resulting in severely insufficient interpretability. These issues hinder the reliable application of prompt engineering in critical medical tasks [9].

Previous studies have laid an important foundation: CLIP has demonstrated the generalization potential of foundational models in zero-shot COVID-19 diagnosis tasks, while BiomedCLIP, as a representative domain-specific model, has achieved state-of-the-art performance on multiple benchmark datasets through fine-tuning on biomedical corpora [4, 10]. However, a significant research gap remains: although model fine-tuning techniques have matured, research on optimizing prompt mechanisms is still in a fragmented stage, especially the lack of a systematic evaluation framework for prompt effectiveness [2].

Based on this, this paper attempts to construct a structured prompt evaluation system: by establishing a multi-dimensional evaluation framework, it provides a quantifiable optimization pathway for prompt design. This framework will enable standardized evaluation and attribution analysis of prompt strategies, fundamentally enhancing the interpretability and clinical credibility of prompt engineering. Pre-trained models optimized through this framework are expected to significantly outperform baseline models in COVID-19 chest X-ray classification tasks.

II. METHODOLOGY

A. Data Preparation

This study utilized a publicly accessible repository containing 21,165 chest X-ray (CXR) images across four diagnostic categories: 3,616 COVID-19-positive cases, 10,192 normal pulmonary findings, 6,012 non-COVID-19 infectious pulmonary opacities, and 1,345 cases of viral pneumonia [1, 11]. Comprehensive preprocessing pipelines were subsequently applied to this dataset. It was divided into three parts: 80% of the images constituted the training set, while the test set and validation set each accounted for 10%. During image preprocessing, the preprocessing pipeline was intentionally set to a minimum, preserving diagnostic information in medical images while also normalizing the input images for the BiomedCLIP model. Research has shown that medical images, particularly chest X-rays, contain subtle textures and small lesions that are diagnostically critical [12]. These details can be lost through aggressive preprocessing, so this study opted for a conservative strategy that prioritizes preserving diagnostic information over aggressive enhancement.

B. The Diagnostic Axis Framework

TABLE I. Prior research has underscored the importance of context-aware prompting, and related studies have similarly demonstrated that subtle differences in the wording of prompts

can significantly influence the zero-shot capabilities of models [2, 9]. The methodology presented in this paper consists of two core components. First, it introduces three distinct diagnostic dimensions (i.e. Abstraction, Certainty and Viewpoint) to systematically examine the model's sensitivity to various prompt styles and how this sensitivity affects its performance. Second, building on the understanding of the model's responsiveness to different prompts, the study hypothesizes that a composite prompt—incorporating the most impactful features identified across all three diagnostic axes—should yield optimal performance.

Axis of Abstraction. This axis delineates the granularity spectrum of prompt formulation, progressing from generalized diagnostic categorizations to specific, empirically observable imaging manifestations. Level A1 embodies the highest abstraction tier, explicitly designating pathological conditions. Level A2 transitions toward physiological characterizations that articulate underlying disease mechanisms. Level A3 represents the least abstract stratum, detailing objective visual patterns detectable within chest radiographs.

Axis of Certainty. It quantifies how strongly a clinical prompt asserts a diagnostic conclusion, assessing the model's sensitivity to different levels of cognitive modality. Level B1 uses definitive assertions that present diagnoses as categorical facts. Level B2 introduces high-confidence qualifiers, such as characteristic manifestations or typical signs, to enhance the reliability of the assessment. Level B3 employs probabilistic language, including suggestive findings and consistent features.

Axis of Viewpoint. To modulates the conceptual perspective used for image characterization, the axis ranges from patient-centered descriptions to formal radiological technical language. Level C1 employs first-person clinical experience terminology, emphasizing symptomatic manifestations described in lay-accessible language. Level C2 adopts a clinician-oriented framework using standardized medical nomenclature. Level C3 represents the most formal stage, utilizing a structured radiological lexicon that incorporates modality-specific reporting conventions.

Generated prompts are enumerated in Table I. Guided by this framework, this study designed corresponding prompts for the four categories, aligning the classification method with the dataset adopted in this work to facilitate experimental validation. The prompt design drew reference from the Centers for Disease Control and Prevention (CDC) guidelines and multiple medical studies capturing the radiological features of COVID-19 to ensure the accuracy and reliability of pathological descriptions and terminology [13-16].

TABLE I. DIAGNOSTIC AXIS FRAMEWORK PROMPT

Class	Axis Level	Prompt
COVID	A1	chest x-ray of COVID-19 pneumonia
	A2	lung inflammation due to SARS-CoV-2 infection
	A3	chest x-ray with peripheral ground-glass opacities and consolidation in both lungs due to COVID-19
	B1	this is an x-ray of COVID-19 pneumonia
	B2	chest x-ray characteristic of COVID-19 pneumonia
	B3	chest x-ray findings are suggestive of COVID-19 pneumonia
	C1	a patient with COVID-19
	C2	chest x-ray showing bilateral ground-glass opacities characteristic of COVID-19 pneumonia
	C3	chest radiograph demonstrating bilateral peripheral airspace opacities characteristic of COVID-19 pneumonia

Normal	A1	chest x-ray of normal lungs
	A2	healthy lung tissue with no inflammation

CONTINUE TABLE I

Class	Axis Level	Prompt
Normal	A3	chest x-ray with clear lung fields, visible lung markings, and no white opacities
	B1	this is a normal chest x-ray
	B2	chest x-ray typical of a healthy individual
	B3	chest x-ray findings are consistent with normal lungs
	C1	a patient with a healthy lung
	C2	chest x-ray showing clear lung fields with no abnormalities or infiltrates
Lung Opacity	C3	unremarkable chest radiograph with no acute cardiopulmonary findings
	A1	chest x-ray of lung opacity with non-COVID infection
	A2	lung inflammation or fluid causing opacity
	A3	chest x-ray with focal white opacities in the lung fields indicating infection or inflammation
	B1	this is an x-ray of lung opacity
	B2	chest x-ray showing clear signs of lung opacity
Viral Pneumonia	B3	chest x-ray findings are suggestive of lung opacity
	C1	a patient with a lung infection
	C2	chest x-ray showing patchy lung opacities consistent with non-COVID bacterial pneumonia
	C3	chest radiograph showing patchy airspace opacities suggestive of bacterial pneumonia
	A1	chest x-ray of viral pneumonia
	A2	lung inflammation due to a viral infection
	A3	chest x-ray with diffuse reticular or reticulonodular pattern throughout both lungs
	B1	this is an x-ray of viral pneumonia
	B2	chest x-ray characteristic of viral pneumonia
	B3	chest x-ray findings are suggestive of viral pneumonia
	C1	a patient with the flu or a virus
	C2	chest x-ray showing interstitial infiltrates typical of viral pneumonia
	C3	chest radiograph with diffuse interstitial infiltrates consistent with viral pneumonia

III. EXPERIMENTS

A. Experimental Setup

For baseline evaluation, the unmodified BiomedCLIP model performed zero-shot predictions on the test set to assess its intrinsic capability. BiomedCLIP is an advanced variant of the CLIP model tailored for the biomedical domain, incorporating improvements over the original CLIP to better align with biomedical image and text characteristics. Key enhancements include replacing the original text encoder GPT-2 with PubMedBERT, employing larger Vision Transformer models, and introducing a patch dropout strategy to improve performance while maintaining pre-training efficiency [10]. Specifically, the model checkpoint combines a Vision Transformer (ViT-Base/16) image encoder with a PubMedBERT text encoder. Input images were standardized to 224×224 pixel resolution, with representations optimized within a shared multimodal embedding space. In the baseline approach, the model receives simple prompts as shown in Table II. In addition, this study applied multiple evaluation metrics, including accuracy, precision, recall, and F1 score.

TABLE II. BASELINE PROMPT TEMPLATES

Class	Prompt
COVID	A patient with COVID-19 pneumonia
Normal	A healthy patient with normal lungs
Lung Opacity	A patient with non-COVID lung opacity
Viral Pneumonia	A patient with viral pneumonia

B. Experimental Results

This section presents a comparative analysis between baseline models and fine-tuned variants, while demonstrating the model's responsiveness to diverse prompt styles under the diagnostic axis framework. Results show that the model achieved optimal in-frame performance when employing prompts that incorporated: direct descriptions of pathological visual patterns, suggestive language constructs, and clinician-oriented instructional phrasing. Building on these findings, a composite prompt integrating these characteristics was formulated (see Table III), which subsequently delivered the highest overall performance. This confirms the theoretical hypothesis of this paper. The original model's overall accuracy is 50%. Compared to it, the fine-tuned model has a 20.33% performance improvement, which is 70.33%.

A comprehensive set of performance metrics is provided in Tables IV and V. In the classification of COVID-19 and Lung Opacity, the model exhibited significant improvements in precision, recall, and F1-score. This shows that the model has been greatly enhanced in both discrimination ability and stability. For the Normal class, although accuracy slightly decreased, both recall and F1-score improved noticeably, indicating enhanced overall stability of the model. Notably, for the Viral Pneumonia category, while precision increased, both recall and F1-score declined significantly. This observation aligns with previous findings that COVID-19 symptoms can be challenging to distinguish from other viral respiratory diseases and may exhibit overlapping features [11]. As a result, the introduction of more refined prompts, contrary to expectations, led to inferior performance in this specific class compared to the baseline model.

Overall, the model optimized through the diagnostic axis framework demonstrated improved overall performance compared to the baseline model in COVID-19 classification tasks. Specifically, the model exhibited a significantly enhanced ability and greater stability in identifying abnormal lung opacities (including both COVID and Lung opacities). In the task of recognizing normal lung cases, the model adopted a more stringent criterion, trading off a certain degree of precision for higher recall and improved robustness. Regarding viral pneumonia classification, the results indicate that more refined prompts contributed to increased precision. However, this also led to a higher number of true positive samples being incorrectly assigned to other categories, resulting in substantially reduced recall and compromised stability in this specific class.

These findings suggest that while targeted prompt design can enhance certain aspects of model performance, it may also introduce trade-offs between precision and recall—particularly in visually or semantically overlapping categories such as viral pneumonia and COVID-19. It highlights the importance of developing task-aware and category-balanced prompt strategies to mitigate such side effects and achieve more consistent performance across all target classes.

TABLE III. COMPOSITED PROMPT TEMPLATES

Class	Prompt
COVID	chest radiograph showing peripheral ground-glass opacities and consolidation in both lungs, findings highly suggestive of COVID-19 pneumonia
Normal	chest radiograph showing clear lung fields and well-defined lung markings, consistent with normal lungs
Lung Opacity	chest radiograph showing focal white opacities in the lung fields, indicating infection or inflammation, suggestive of lung opacity
Viral Pneumonia	chest radiograph showing diffuse reticular or reticulonodular pattern throughout both lungs, suggestive of viral pneumonia

TABLE IV. BASELINE PERFORMANCE ON ZERO-SHOT PREDICTION

Class	Evaluation Metric	Value
COVID	Precision	19.94%
	Recall	18.73%
	F1-Score	19.32%
Normal	Precision	83.13%
	Recall	74.90%
	F1-Score	78.80%
Lung Opacity	Precision	30.71%
	Recall	25.25%
	F1-Score	27.71%
Viral Pneumonia	Precision	20.82%
	Recall	56.30%
	F1-Score	30.40%

TABLE V. FINE-TUNING PERFORMANCE ON ZERO-SHOT PREDICTION

Class	Evaluation Metric	Value
COVID	Precision	47.11%
	Recall	31.40%
	F1-Score	37.69%
Normal	Precision	76.89%
	Recall	94.90%
	F1-Score	84.95%
Lung Opacity	Precision	69.35%
	Recall	64.29%

Viral Pneumonia	F1-Score	66.72%
	Precision	36.07%
	Recall	16.30%
	F1-Score	22.45%

C. Limitations and Future Work

One notable limitation of this study lies in its oversight of potential interactions among prompts of varying styles, such as inherent correlations or conflicts between different prompt designs. This neglect may result in the redundancy or omission of critical information, which could otherwise contribute to further improving the model's classification accuracy. A further limitation is the framework's limited support for complex multi-step reasoning. Its success in COVID-19 image classification may not generalize when prompts grow more elaborate across multiple conceptual axes. In such cases, advanced methods (such as chain-of-thought prompting) may be necessary to sustain coherent reasoning and performance [2].

Future work will analyze how different prompt types interact and add interaction aware mechanisms to the framework to enable more effective composite prompts with less interference. Evaluation will extend to complex, multi step clinical decision contexts. These steps aim to limit information loss and misinterpretation and to strengthen real world reliability.

IV. CONCLUSION

This work fine-tunes a pre-trained CLIP model for COVID-19 image classification and shows, through systematic experiments, that structured prompts markedly improve performance. The method exploits pre-trained generalization while strengthening semantic alignment via designed templates. Comparative prompt studies reveal pronounced sensitivity to wording and structure, motivating the standardization of prompt construction. Several limitations persist, however, such as insufficient exploration of interactions among prompt styles and limited support for complex reasoning tasks. Future research should account for potential inter-prompt interactions and extend the framework to accommodate multi-step reasoning. In summary, this work provides an efficient adaptation strategy for vision-language models in downstream applications that circumvents the computational expense of large-scale retraining, particularly medical image classification. It effectively leverages the transfer capabilities of pre-trained models, suggesting a viable technical direction for improving their generalizability and practical deployment in real-world contexts.

REFERENCES

- [1] M. E. H. Chowdhury, T. Rahman, A. Khandakar, R. Mazhar, M. A. Kadir, Z. B. Mahub et al., "Can AI Help in Screening Viral and COVID-19 Pneumonia?," IEEE Access, vol. 8, pp. 132665–132676, 2020.
- [2] H. Yang, X. L. Liang, Z. Li, Y. Sun, Z. Y. Hu, X. H. Xie et al., "Prompt Mechanisms in Medical Imaging: A Comprehensive Survey," 2025, arXiv:2507.01055.
- [3] H. N. Lin, C. Xu and J. Qin, "Taming Vision-Language Models for Medical Image Analysis: A Comprehensive Review," 2025, arXiv:2506.18378.

- [4] Z. Zhao, Y. X. Liu, H. Wu, M. Wang, Y. H. Li, S. Wang et al., "CLIP in medical imaging: A survey," *Medical Image Analysis*, vol. 102, p. 103551, Mar. 2025.
- [5] S.-C. Huang, L. Shen, M. P. Lungren, and S. Yeung, "GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-efficient Medical Image Recognition," Oct. 2021.
- [6] Z. Wang, Z. Wu, D. C. Agarwal, and J. Sun, "MedCLIP: Contrastive Learning from Unpaired Medical Images and Text," *arXiv (Cornell University)*, Jan. 2022.
- [7] M. Guo, H. Yi, Z. Qin, H. Wang, A. Men, and Q. Lao, "Multiple Prompt Fusion for Zero-Shot Lesion Detection Using Vision-Language Models," *Lecture notes in computer science*, pp. 283–292, Jan. 2023.
- [8] J. Silva-Rodríguez, Hadi Chakor, Riadh Kobbi, J. Dolz, and Ismail Ben Ayed, "A Foundation Language-Image Model of the Retina (FLAIR): Encoding expert knowledge in text supervision," *Medical Image Analysis*, vol. 99, pp. 103357–103357, Oct. 2024.
- [9] A. Ahmed, M. Hou, R. Xi, X. Zeng, and Syed Attique Shah, "Prompt-Eng: Healthcare Prompt Engineering: Revolutionizing Healthcare Applications with Precision Prompts," May 2024.
- [10] S. Zhang, Y. B. Xu, N. Usuyama, H. W. Xu, J. Bagga, R. Tinn et al., "A Multimodal Biomedical Foundation Model Trained from Fifteen Million Image–Text Pairs," *NEJM AI*, vol. 2, no. 1, Dec. 2024.
- [11] T. Rahman, A. Khandakar, Y. Qiblawey, A. Tahir, S. Kiranyaz, S. B. A. Kashem et al., "Exploring the effect of image enhancement techniques on COVID-19 detection using chest X-ray images," *Computers in Biology and Medicine*, vol. 132, p. 104319, May 2021.
- [12] Pär Kragsterman, "The Ultimate Guide to Preprocessing Medical Images: Techniques, Tools, and Best Practices for Enhanced Diagnosis," *Collectiveminds.health*, Aug. 28, 2024. <https://collectiveminds.health/articles/the-ultimate-guide-to-preprocessing-medical-images-techniques-tools-and-best-practices-for-enhanced-diagnosis>
- [13] CDC, "Clinical Presentation," COVID-19, 2024. <https://www.cdc.gov/covid/hcp/clinical-care/covid19-presentation.html>
- [14] S. Nakakubo, "Evolving COVID-19 symptoms and the ongoing course of research," *The Lancet Infectious Diseases*, vol. 25, no. 3, pp. 245–246, Mar. 2025.
- [15] Z. Sadiq, S. Rana, Z. Mahfoud, and A. Raoof, "Systematic review and meta-analysis of chest radiograph (CXR) findings in COVID-19," *Clinical Imaging*, vol. 80, pp. 229–238, Dec. 2021.
- [16] J. P. Kanne, H. Bai, A. Bernheim, M. Chung, L. B. Haramati, D. F. Kallmes et al., "COVID-19 Imaging: What We Know Now and What Remains Unknown," *Radiology*, p. 204522, Feb. 2021.